

Ish Shah

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Research interests

Harmonic analysis and its interactions with related fields, including analytic number theory and partial differential equations.

Education

- 2026–present **Ph.D. in Mathematics**, University of Illinois Urbana-Champaign.
2022–2026 **B.S. in Mathematics** (highest honors), Rutgers University–New Brunswick.
Minor in Computer Science. GPA: 4.0/4.0.

Papers

- Publications **Shimura lifts of nearly holomorphic modular forms.**
Joint with Abby Linscott, Tianyu Ni, and Hui Xue.
Internat. J. Math., 37(07), 2026.
- Pointwise ergodic theorems along fractional powers of primes.**
Joint with Erik Bahnson, Leonidas Daskalakis, and Abbas Dohadwala.
Int. Math. Res. Not. IMRN, 2025(15), 2025. (arXiv:2412.07055)

Research talks and presentations

- Sep. 2025 *Shimura lifts of almost holomorphic modular forms.*
Number Theory Seminar, Rutgers University–New Brunswick.
- Jul. 2025 *A Selberg identity for nearly holomorphic modular forms.*
12th Annual Summer Undergraduate Research Symposium, Clemson University.
- Jul. 2024 *When Fourier analysis meets ergodic theory and number theory.*
DIMACS REU, Rutgers University–New Brunswick.

Awards

- Jun. 2026 **Matthew Leydt Society Inductee** (Rutgers University–New Brunswick).
Inducted to a society for students with “truly exceptional” performance, roughly constituting the top 2% of the class.
- Apr. 2026 **Graduate Research Fellowship Program Honorable Mention** (National Science Foundation).
Awarded an Honorable Mention in the national NSF fellowship competition.
- Apr. 2026 **Kenneth and Rosalind Wolfson Annual Award for Academic Excellence in Mathematics** (Department of Mathematics, Rutgers University).
Awarded at graduation to a Mathematics major, based on academic merit.
- Nov. 2025 **Alan Marc Schreiber Memorial Scholarship** (School of Arts and Sciences, Rutgers University). Also awarded in Jan. 2025 and Feb. 2024.
Awarded to Mathematics and Economics students, based on academic merit.

- Sep. 2025 **Maurice M. and Adrienne R. Weill Scholarship** (Department of Mathematics, Rutgers University). Also awarded in Aug. 2024.
Awarded to Mathematics majors, based on academic merit.
- Jan. 2025 **Rutgers College Scholarship** (School of Arts and Sciences, Rutgers University).
Based on academic merit.
- Dec. 2024 **Goldwater Scholarship Nomination** (Office of Distinguished Fellowships, Rutgers University).
One of four nominees from Rutgers–New Brunswick selected to advance to the national competition, determined following a competitive application process.
- Sep. 2024 **Excellent TA/PTL/Grader Award** (Department of Computer Science, Rutgers University).
Awarded to four undergraduate students and several graduate students based on reviews from faculty.

Service and outreach

- 2025–2026 **President**, Rutgers Undergraduate Mathematics Association.
- 2025–2026 **Lecturer**, Rutgers Competitive Programming.
- 2026 **Workshop Instructor**, Rutgers University Security Club.
- 2024–2025 **Public Relations Officer**, Rutgers Undergraduate Mathematics Association.

Teaching

Any roles below this line were completed at Rutgers University as an undergraduate student.

- Spring 2026 **CS 344 (Algorithms)**, Part-Time Lecturer/Teaching Assistant.
CS 112 (Data Structures), Learning Assistant.
- Fall 2025 **CS 111 (Introduction to Computer Science)**, Learning Assistant.
- Spring 2025 **CS 344 (Algorithms)**, Part-Time Lecturer/Teaching Assistant.
Math 152 (Calculus II), Learning Assistant.
- Fall 2024 **CS 344 (Algorithms)**, Part-Time Lecturer/Teaching Assistant.
Math 152 (Calculus II), Learning Assistant.
- Spring 2024 **CS 344 (Algorithms)**, Part-Time Lecturer/Teaching Assistant.
CS 112 (Data Structures), Learning Assistant.
- Fall 2023 **BAIT 370 (Management Information Systems)**, Learning Assistant.